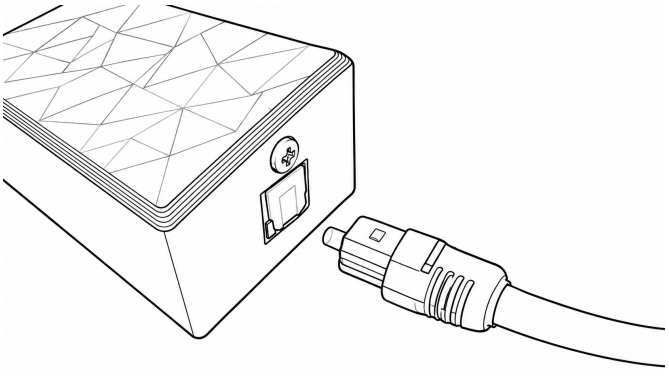
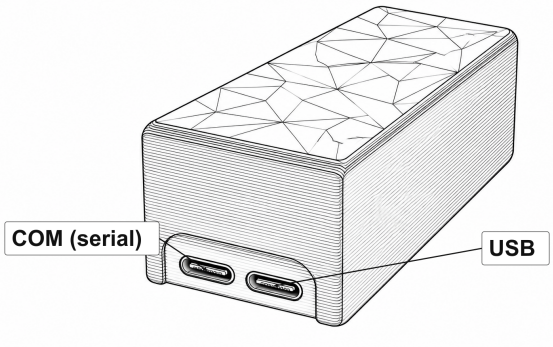
Optical Audio Bridge

Setup guide

For an optical-input amplifier or soundbar, controlled with Home Assistant.



Connecting the optical cable



Pictured unit: left COM, right USB

1. Connect the device

Optical: Remove the protective caps. Plug the TOSLINK cable into the bridge and connect the other end to the optical input (OPTICAL IN) on your sound system. Align the plug and insert it without forcing it.

Power: Connect a 5 V USB power supply to one USB-C port. On the pictured unit, the left port is USB-to-serial (COM) and the right port is native USB.

Turn on your sound system, select its optical input and set a moderate volume. On the tested JBL soundbar, the input is called **TV**.

2. Set up Wi-Fi

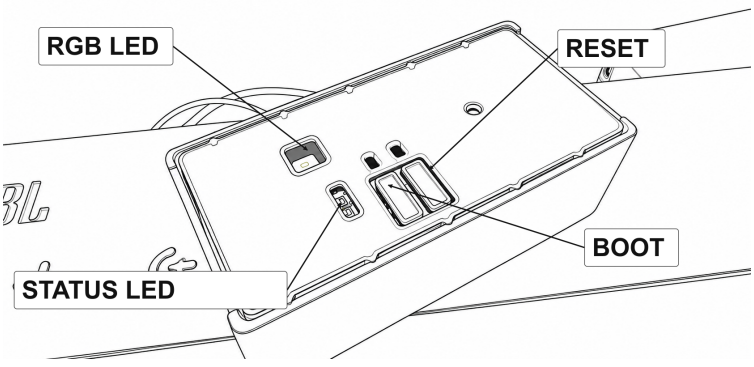
If your Wi-Fi details were configured in advance, the device will try to connect automatically when powered on; you can skip this step.

Otherwise, wait about 90 seconds and connect your phone to the open **optical-audio-bridge-...** Wi-Fi network. Your phone may report that there is no internet.

Open the setup page, or enter **http://192.168.4.1** in your browser. Select your **2.4 GHz** Wi-Fi, enter its password and save. Wait 30-60 seconds, then reconnect your phone to your usual Wi-Fi.

Saved Wi-Fi details survive power loss. The setup network may reappear when Wi-Fi is unavailable; this does not erase the saved details.

3. Buttons and indicators



RESET: restarts the device without erasing Wi-Fi settings.

BOOT: used for programming; not needed in everyday use.

RGB LED: colour and brightness are adjustable in Home Assistant.

4. Add to Home Assistant

Connect Home Assistant and the bridge to the same home network. In Home Assistant, open **Settings → Devices & services**.

For the discovered ESPHome device, choose **Configure** and complete the setup. If it is not discovered, choose **Add integration → ESPHome** and enter the device IP address. No API key is needed with the default firmware. You can then assign the device to a room and rename it.

5. Music, speech and the LED

Home Assistant provides an **Optical Audio Bridge** media player and an RGB light. You can play supported media or text-to-speech announcements on the player, and change the light colour and brightness.

For music services, you can install **Music Assistant**. In its provider settings, enable **Home Assistant Media Players** and select the Optical Audio Bridge player. Set up your preferred music provider separately.

The YouTube Music Cast button and the direct Windows Wi-Fi audio output do not discover this device. Home Assistant media does not require Music Assistant.

6. ESPHome and network access

The device runs ESPHome; everyday Home Assistant use does not require opening ESPHome Device Builder. Device Builder may offer **Take control** later, but adoption and its first update have not been tested.

Security: The setup Wi-Fi and its firmware-upload portal are open. The ESPHome API is unencrypted and wireless updates have no password by default. Anyone with network access can control or update the device; use a trusted network. API encryption and an OTA password can be added later.

If there is no sound or the device is missing

Check the optical cable, sound system input and volume. If Home Assistant cannot find the device, check the 2.4 GHz Wi-Fi and that both are on the same network.

Thank you for your trust. I hope you enjoy using the device!

Questions? Contact hello@davidmolnar.dev or visit the GitHub project.

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